How Interestingness Measures Shape Association Rule Mining

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All figures use colour-blind friendly palettes and are accompanied by descriptive captions

1. Introduction

The technique of finding patterns in data that occur together is called association rule mining, or ARM. When given a collection of transactional data, ARM can identify patterns such as "customers who buy A tend to buy B." ARM has been used in genomics, web clickstream analysis, and medical diagnosis, although its main application is in retail recommendation systems.
Three metrics are introduced in the standard explanation of ARM: lift (the probability of A and B co-occurring more than by chance), confidence (the probability of B given A), and support (the probability of A and B co-occurring). We also present new measures in this tutorial that are more effective in specific situations. The choice of metric fundamentally changes the results of the ARM task, and choosing confidence or lift as the primary metric can result in incorrect conclusions about the data at times.

Objective: By the end of this tutorial, we will be able to explain why standard measures fail, calculate different interestingness measures, and pick the appropriate measure for your data.

2. Rules, Frequency, and the Standard Measures

Agrawal and Srikant (1994) introduced the problem of mining association rules from large transaction databases and proposed Apriori algorithm. A rule $A \rightarrow B$ is characterized by three core quantities:

Measure	Formula	Interpretation
Support	$P(A \cup B)$	Fraction of transactions containing both A and B
Confidence	$P(A \cup B) / P(A)$	Fraction of transactions that also contain B
Lift	$P(A \cup B) / (P(A) \cdot P(B))$	Ratio of observed to expected co occurrence, >1 means positive association

Table 1: Three standard measures in ARM.

Frequent itemsets, which are collections of items whose support is higher than the minimum support value, are used to create candidate rules. We employ the FP Growth approach to effectively identify frequent itemsets (Han et al., 2000). This approach saves a significant amount of memory by storing the database in a compressed form as a prefix tree structure known as the FP tree, whereas the Apriori method repeatedly scans the database.

2a. Example

Before we get into the choice of measures, we go through the pipeline of ARM on a small example. This is to make sure that we see each quantity before we use it in the full data set.

The dataset

Imagine a small bakery with 8 customer transactions. Each row is one basket:

Transaction	Items purchased
T1	Bread, Butter, Milk
T2	Bread, Butter
T3	Bread, Milk
T4	Bread, Butter, Jam
T5	Butter, Milk
T6	Bread, Jam
T7	Bread, Butter
T8	Milk, Jam

Table 2: Eight bakery transactions used in the worked example.

Step 1: Count item frequencies

With N = 8 transactions, count how many times each item appears:

Item	Count	Support P(item)
Bread	6	6/8 = 0.75
Butter	5	5/8 = 0.625
Milk	4	4/8 = 0.50
Jam	3	3/8 = 0.375

Table 3: Item frequencies and support values (min support threshold = 0.50).

Setting minimum support threshold of 0.50, all four items are frequent. Checking pairs: {Bread, Butter} appears in T1, T2, T4, T7 → support = 4/8 = 0.50. {Bread, Milk} appears in T1, T3 → support = 2/8 = 0.25 (below threshold). Only {Bread, Butter} has support higher than the threshold.

Steps 2 and 3 : Generate rules and evaluate

From the frequent itemset {Bread, Butter} we generate the following candidate rules:

Rule	Support	Confidence	Lift	Cosine	Kulczynski	Conviction
Bread →Butter	0.50	0.667	1.07	0.73	0.733	1.125
Butter →Bread	0.50	0.800	1.07	0.73	0.733	1.250

Table 4: Both candidate rules from {Bread, Butter} itemset.

From the exmaple, we get the following conclusions : Confidence is not symmetric, even though lift is identical. This reflects that bread is purchased more broadly (75% of baskets) while butter

buyers are more selective. The kulczynski is 0.733 which is > 0.5 which means that the association is positive and both kulczynski and cosine are null invarient.

3. The Problem with Confidence: Null Transactions

Tan, Kumar, and Srivastava (2004) discovered a crucial flaw in the concept of lift and confidence, which is nearly always overlooked in basic introductions: it is not a null invariant metric.
A transaction that lacks both A and B is called a null transaction. It is incorrect to assume that any two products will appear in every transaction as a combination in a supermarket with thousands of products; rather, they will be jointly absent in the great majority of transactions. Even though the actual relationship between A and B has not changed, the values of lift and confidence will as the number of null transactions increases.
Let's look at two distinct scenarios with exactly the same number of A and B combinations. There are 800 null transactions in the first scenario and 8,800 in the second. The lift increases from 1.88 to 4.55, but the real relationship between A and B stays the same. The impact on all of the fundamental metrics is displayed in Figure 1.

Consequently, a measure is considered null invariant if the addition or removal of null transactions has no effect on its value. The null invariant property is not satisfied by lift and confidence measures. This is crucial for large sparse data applications like text mining, genomics, and e-commerce.

4. Five Interestingness Measures

We examine five measures. Two (Kulczynski and Cosine) are null invariant and thus more appropriate when items are rare relative to the full catalogue.

Measure	Formula	Range	Null-invariant?	Weakness
Lift	$P(A \cup B) / (P(A) \cdot P(B))$	$[0,\infty)$ 1=indep.	**No**	Inflated by null transactions
Leverage	$P(A \cup B) - P(A) \cdot P(B)$	$[-1,1]$ 0=indep.	**No**	Biased toward frequent items
Conviction	$(1-P(B)) / (1-conf)$	$[0,\infty)$ 1=indep.	**No**	Undefined when conf = 1
Kulczynski	$0.5 \cdot (P(B\|A) + P(A\|B))$	[0,1] 1=perfect	**Yes**	Less intuitive than lift
Cosine (IS)	$P(A \cup B) / \sqrt{(P(A) \cdot P(B))}$	[0,1] 1=perfect	**Yes**	Symmetric; loses directionality

Table 5: Five interestingness measures

The Kulczynski measure (Tan et al, 2004) is especially appropriate for sparse data, as it takes the average of the two conditional probabilities P(B|A) and P(A|B) in a symmetric manner, without dependence on the number of transactions. A Kulczynski measure higher than 0.5 indicates that the items co occur more often than they appear individually. The Cosine measure, also called the IS measure, is the geometric mean of the same two values, preferring rules where items have similar base rates.

5. Empirical Comparison on Real Retail Data

We use the approaches to the Online Retail II data set (available from the UCI Machine Learning Repository; Chen et al., 2012), which includes all transactions for a UK based online retail shop from 2009 to 2010. Limiting to UK transactions and the 100 most common items yields a good set of rules with FP-Growth with min supp = 0.02 and min conf = 0.3.

Figure 1: Effect of Null Transactions on Interestingness Measures

A log-scale representation of the three non-null invariant measures is shown on the left side of this figure. Lift rises sharply from 1.2 to 6.0, while Confidence rises slightly and Leverage gets closer to zero, indicating that these measures react to the quantity of unrelated transactions included. The Cosine and Kulczynski measures are represented linearly on the right side, with both bars appearing to be the same height for both scenarios. This shows that the measures do not react to the addition of null transactions, which makes them a valid representation of the real A and B relationship. It also illustrates how various measures of interestingness react to null transactions.

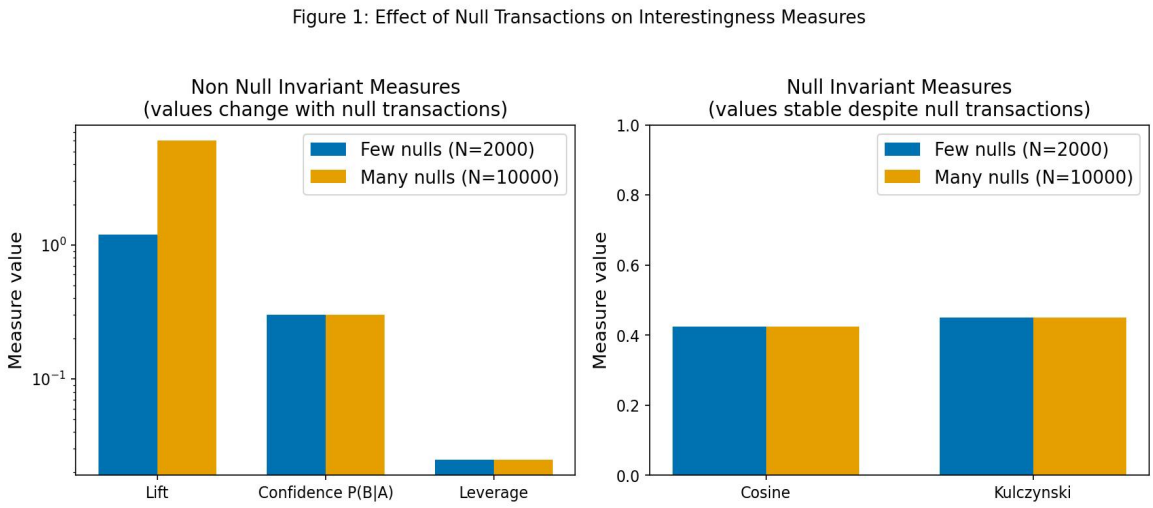


Figure 2: Rank Correlation Heatmap

We compute Spearman correlations between all five measures over the entire set of rules. The heatmap shows that lift and leverage are highly correlated, which is as expected from their given definitions. Kulczynski and Cosine form a second cluster. Conviction is the least correlated with the other measures and is the measure least likely to agree with the others. Using lift when conviction is more appropriate will yield different results.

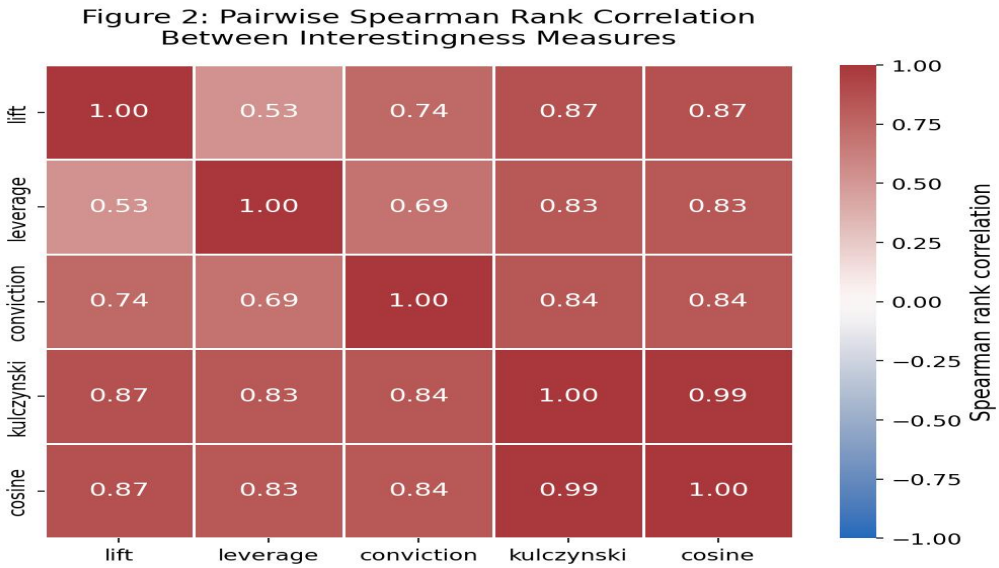


Figure 3: Lift vs Kulczynski Scatter Plot

All of the rules are shown in Figure 3 based on their lift values on the x-axis and their Kulczynski score values on the y-axis, with color coding indicating whether or not they rank in the top 20 for each measure. The variation, which indicates that rules with low lift have high Kulczynski and those with high lift have low Kulczynski, is an important finding. When one item is common and the other is rare, the lift is skewed by the rare item instead of the fact that they go together. This is why the former have high lift and low Kulczynski.

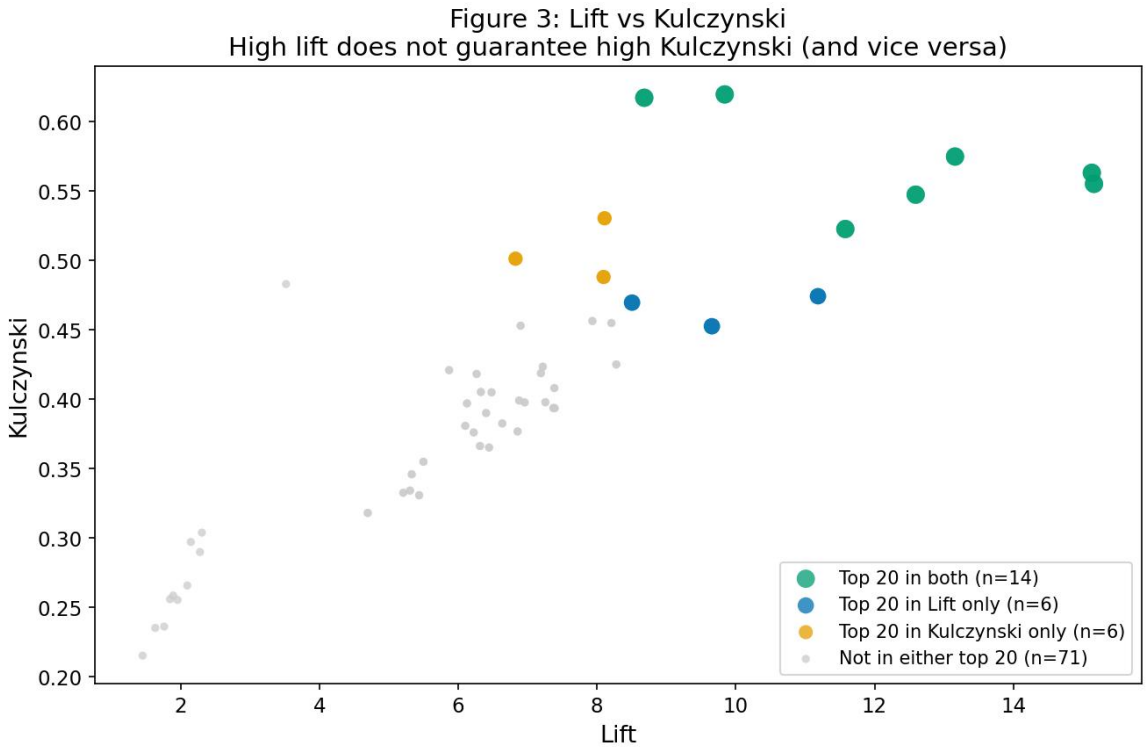
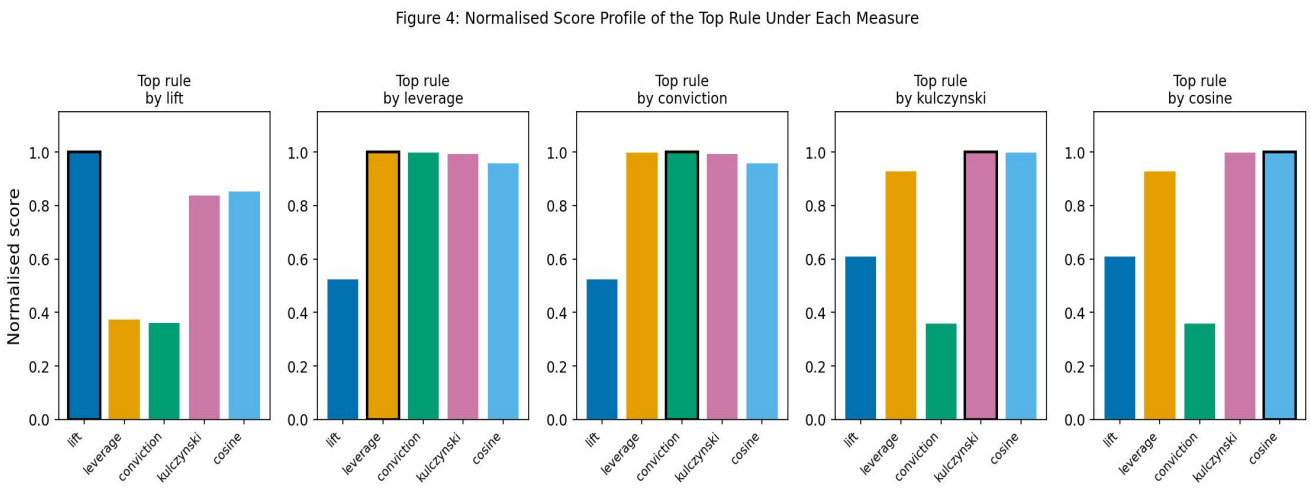


Figure 4: Score Profiles of Top Rules

For each measure, the best rule is identified and the normalised score profile is plotted across all five measures. This shows that the 'best' rule according to the lift metric may have a low conviction score, and vice versa. It is more important for a rule to perform well across many measures than to perform well on a single measure .



Across all five measures on this dataset, the top 20 rule sets overlap by only 40–60%. You would surface materially different business recommendations depending purely on your choice of measure.

6. When to use which measure?

Situation	Recommended Measure	Reason
Large catalogue, rare items (e-commerce, genomics)	**Kulczynski or Cosine**	Null invariant, stable despite catalogue size
Dense, balanced transactions (supermarket loyalty)	**Lift or Leverage**	Intuitive, widely understood by stakeholders
Need directional rule strength (recommendation)	**Conviction**	Penalises rules where B is nearly certain regardless of A
Unknown or mixed data properties	**Compute all five and compare**	Disagreement signals structure worth investigating

Table 6: When to use what measures

Mining rules once and then evaluating them simultaneously using several criteria is a good approach. Rules that show up in the top K results across several criteria are more likely to be true patterns. However, if a rule appears to be highly sensitive to a single dataset, it is important to consider the possibility that the situation is just very sensitive to the rule.In their extensive review of more than thirty interestingness criteria, Geng & Hamilton (2006) propose that the most important factor for sparse data is null invariance. The findings here support the suggestion that null invariant criteria be used by default when items are sparse in relation to the transaction space.

6. Conclusion

This tutorial demonstrates how difficult it is to choose interestingness metrics in association rule mining. The most widely taught interestingness metrics, confidence and lift, are not null invariant and may produce inaccurate results in sparse or unbalanced data. When the items are not common or even rare in comparison to the entire transaction space, Kulczynski-based interestingness measures may be more appropriate for ranking.
In conclusion, it is advised to compare results using a variety of interestingness metrics, with the ultimate choice based on the data set's statistical characteristics. While agreements merely validate the results, disagreements between interestingness measures suggest very interesting outcomes.

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